Physical Intelligence – D2 Version

BOTTLENECK ANALYSIS

The absence of production-certifiable runtime at the system layer, where hardware and middleware meet, prevents deterministic integration, safety validation, and the ability to scale robot platforms



Deployments stall when robots must run deterministically and safely in brownfields or harsh environments. At the application layer control software must drive physical robots safely while staying tightly integrated with existing systems. Hardware interoperability issues affect 36% of manufacturers, pointing to failures at the middleware layer

Overview

- Bottlenecks force teams to rebuild integration and safety cases for every robot and site
- Healthcare and space face added constraints, requiring regulatory approval (FDA, IEC 62304) and rigorous hardware validation (radiation, vibration, thermal testing) on top of core platform requirements
- Middleware constraint is a productization gap, not a pure research gap: ROS2, real-time Linux, and DDS can achieve near-deterministic lab performance, but are fragile to configure, debug, and certify at scale, so each team still rebuilds a bespoke stack for every robot and site

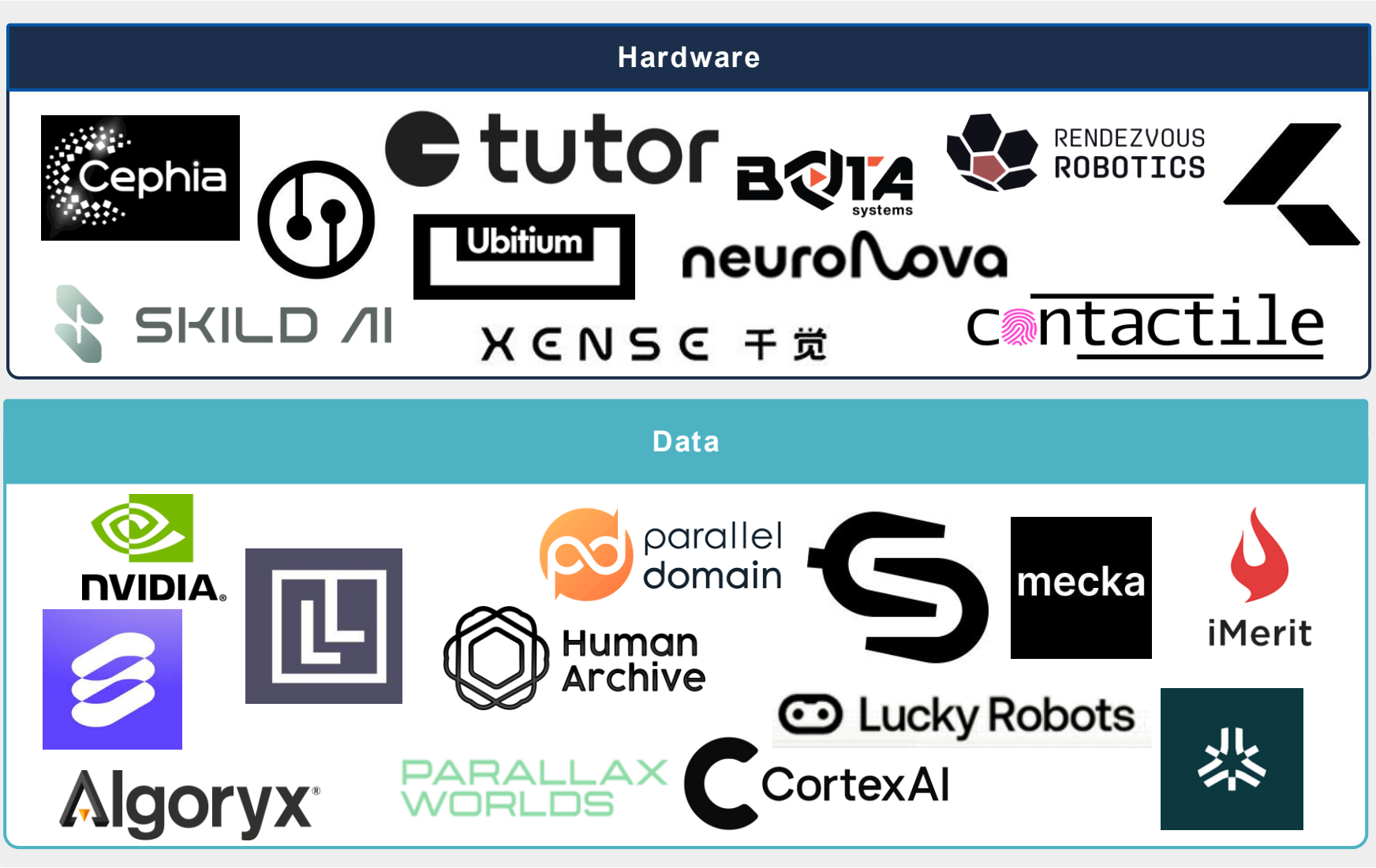
Opportunities

- Middleware today has a relatively small, verticalized TAM and faces strategic pressure from hyperscalers and AI labs that are building pieces of the stack
- A durable business needs to own the safety-critical runtime and the surrounding compliance infrastructure, bundling execution, tooling, and reusable certification. Cloud providers focus up-stack and leave the middleware market underserved
- Thus, the core missing piece is a “robot OS” that allows developers to deploy and scale robots without customizing design and integration for each project

Sources: [1] a16 News [2] Eureka [3] Market Intel [4] Parasoft [5] Robotics Tomorrow [6] Ros Industrial [7] Sigbed [8] SQ Magazine

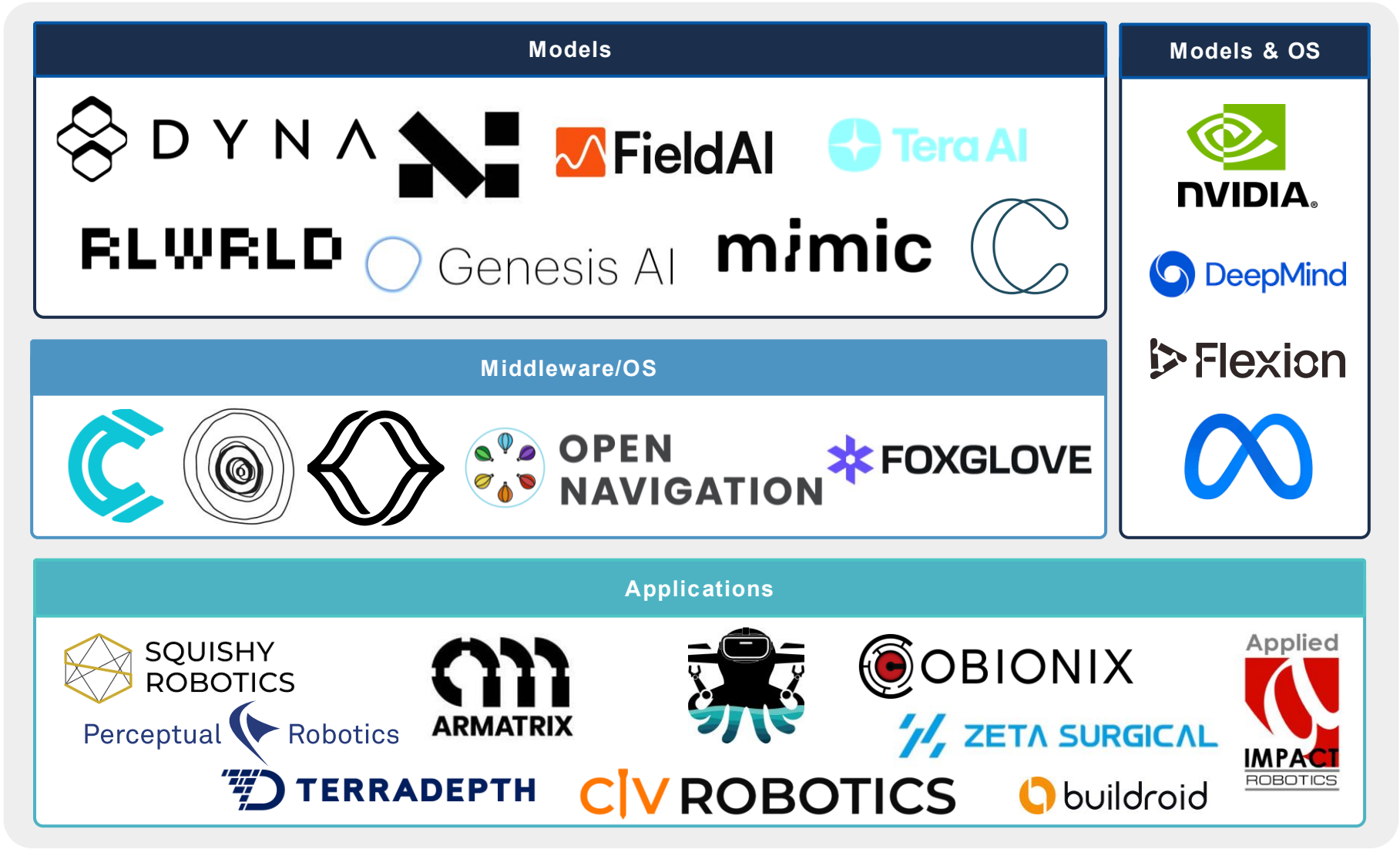
MARKET MAP (I)

Density is highest in data and models, but middleware and hardware remain underserved relative to their role in production deployment, with middleware the clearest bottleneck and the fewest startups building against it



MARKET MAP (II)

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RENDEZVOUS ROBOTICS

Rendezvous Robotics eliminates the need for space-bound robotic hardware to fold before launch by developing manufacturing tiles that launch in stacks and assemble into structures via software commands, powered by autonomous swarm assembly and electromagnetism

Overview

- **Sector:** Space Manufacturing Infrastructure
- **Headquarters:** Golden, CO
- **Year Founded**: 2024
- **Latest Funding Round**: Early Stage VC
- **Amount Raised:** $5.20mm
- **Total Capital Raised**: $8.20mm
- **Number of Employees**: 4
- Rendezvous Robotics develops modular robotics systems that can accomplish diverse amounts of tasks in autonomous in-space assembly, commercializing their technology TESSERAE, or Tessellated Electromagnetic Space Structures for the Exploration of Reconfigurable, Adaptive Environments, as flat-packed tiles that launch in stacks and latch together to form different structures in orbit, commanded by software to unlatch and rearrange when mission parameters change

Investment Thesis

- The primary bottleneck in the deploying orbital infrastructure is packing inefficiency due to fairing limitations. Caltech JPL and NASA research shows that building space infrastructure using “single-launch” architecture is not economically viable, an issue only solved by transitioning to modular assembly, decoupling the structure’s final size from the fairing’s diameter
  - TESSERAE technology solves exactly this. Modular tiles can achieve volumetric efficiency in packing, and the robotic swarm function eliminates the use of robotic assembly arms or human crew
- Dr. Ariel Ekblaw, developer of TESSERAE tech during her time at MIT, is also the founder of Rendezvous. With the team and technology essentially coming out of MIT, Rendezvous is the primary vehicle for the commercial IP and essentially act as “first-movers” in this space. In January of 2026, the technology was proved viable aboard the ISS, successfully utilizing a large visual sensing network and multiple tiles.

Fundraising History

Investors	Capital Raised	Date Raised
Undisclosed	$5.20mm	Mar 2026
Aurelia Foundry, 8090 Industries, ATX Venture Partners, Mana Ventures	$3.00mm	Sep 2025

Key Management

Name	Position	Qualifications
Phil Frank	Co-Founder; CEO	BS in Astronautical Engineering @ University of Michigan
Joe Landon	Co-Founder; President	VP of Advanced Programs Development @ Lockheed Martin, CEO @ Crescent Space

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Sources: [1] Pitchbook [2] LinkedIn [3] Rendezvous Robotics , [4] New Space Economy [5] Caltech JPL [6] DevX [7] Currently in Space [8] MIT Media Lab

SENSEI ROBOTICS

Sensei Robotics aims to allow efficient, low-cost collection of custom robotics datasets from human operators, democratizing access to data collection capacity. Using a gig work model, they ensure rapid data generation without the need for dedicated, full-time teleoperators.

Overview

- **Sector:** Training Data Services
- **Headquarters:** San Francisco, CA
- **Year Founded**: 2024
- **Latest Funding Round**: Early Stage VC
- **Amount Raised:** Undisclosed
- **Total Capital Raised**: $0.50mm
- **Number of Employees**: 5
- Sensei Robotics provides a data collection platform that connects robotics companies to human operators, facilitating the collection of in-the-wild datasets that can be used for robot model training and validation. Human operators act as gig workers, completing data generation requests created by robotics firms. Sensei Robotics sees themselves as the Scale AI for robotics data, creating a social network of robotics teleoperators, researchers, and developers

Investment Thesis

- Sensei is pivotal to the robotics data landscape, operating a decentralized network of human operators (Senseis) who collect real-world demonstration data at a fraction of the cost of traditional teleoperation setups, providing a structural advantage compared to traditional in-house teleoperation
- Sensei Robotics is positioned for growth as a low-cost, high-efficiency, and necessary provider of real-world data as humanoid robotics moves into the real-world data phase in the training pipeline
- Due to its decentralized model that uses individual gig work for training data, Sensei's platform becomes exponentially more powerful with scale. With each new Sensei, the platform becomes more useful to clients as it becomes the one-stop shop for large volumes of diverse and high-quality data
- Sensei holds a distinct cost advantage in its teleoperation setup at $300 a platform, giving it a unique advantage over traditional setups, which have historically cost closer to $40,000

Fundraising History

Investors	Capital Raised	Date Raised
Calibrate Ventures, Lotus Venture Partners, Network VC	Undisclosed	Dec 2024
Y Combinator	$0.50mm	Jul 2024

Key Management

Name	Position	Qualifications
Anubhav Guha	Co-Founder; CEO	Engineer and Program Manager at Aurora Flight Sciences
John Piotti	Co-Founder; CTO	Robotics Engineer at Aurora Flight Sciences

Sources: [1] Pitchbook [2] LinkedIn [3] Rendezvous Robotics

INORBIT.AI

InOrbit captures an unsolved commercial gap in the middleware layer: multi-vendor fleet coordination by building a production-scale infrastructure that ROS2 does not provide. Their Space Intelligence platform sits above any vendor's ecosystem as a hardware-agnostic orchestration layer

Overview

- **Sector:** Cloud-based Robot Orchestration
- **Headquarters:** Mountain View, CA
- **Year Founded**: 2017
- **Latest Funding Round**: Series A
- **Amount Raised:** $10mm
- **Total Capital Raised**: $13.5mm
- **Number of Employees**: 45
- InOrbit is a cloud-based fleet management and robot operations platform that coordinates a mixed fleet of AMRs, AGVs, and drones through their AI-powered product suite, Space Intelligence. They are enabling at-scale adoption of automation across industries to overcome robotics market fragmentation, rigid enterprise software, and isolated physical infrastructure. InOrbit's software-defined orchestration integrates agentic and physical AI-driven insights and real-time monitoring to enhance existing operational systems

Investment Thesis

- No open-source project addresses the production-scale challenge of sustaining systems in which individual components fail, behave unexpectedly, or come from different vendors. Our middleware thesis is that whoever is able to create a commercial layer on top of the open-source standard creates defensible value. InOrbit acts on this opportunity by offering **Space Intelligence**: an advanced "software-defined" orchestration" that the other open-source communities have no incentive to build
- In 2026, InOrbit launched **OpenRobOps**, an open-source fleet management standard. Open-sourcing the core to drive adoption allows InOrbit to capture value in the commercial layer above the open-source foundation
- Enterprise customers, including Colgate-Palmolive and Genentech, confirm that the fleet orchestration market is already spending at production scale

Fundraising History

Investors	Capital Raised	Date Raised
L'ATTITUDE Ventures, Kärcher New Venture, Globant Ventures	$10mm	Sep 2025
ANIMO Ventures	$2.65mm	Jun 2020

Key Management

Name	Position	Qualifications
Florian Pestoni	Co-Founder; CEO	Venture Scout at Alumni Ventures, Founder of Robot Operations Group
Ramiro Diaz Trepat	CTO	Head of Product at xLabs, VP at JPMC

TERRANOVA

Terranova deploys AI-guided autonomous robots to permanently lift flood-prone land through subsurface injection operating in waterlogged, unstable, and otherwise hazardous terrain where traditional construction equipment cannot safely work.

Overview

- **Sector:** AI-Robotics / Flood Adaptation / Climate Tech
- **Headquarters:** Berkeley, CA
- **Year Founded**: 2021
- **Latest Funding Round**: Seed
- **Amount Raised:** $7M
- **Total Capital Raised**: $7M
- **Number of Employees**: <20
- Terranova's robots lift up land out of flood zones using AI-optimized injection patterns and a city-planner interface, used for infrastructure, defense, and construction. The system is targeted at coastal municipalities where seawall costs are prohibitive, quoting $92M to lift 240 acres vs. $900M for traditional seawalls in San Rafael, CA.

Investment Thesis

- Terranova's terraforming robots can be cheaper than levees or seawalls at scale, avoiding heavy construction and surface disruption with faster deployment. It's system quoted $92M to lift 240 acres vs $900M for traditional seawalls in CA, positioning Terranova as a viable solution to flooding
- Terranova's advantage goes beyond the hardware, as the company combines site-suitability prediction, AI/ML planning, and closed-loop control of autonomous injection rovers. This makes deployment efficient and easier to replicate across sites, creating a moat that strengthens as Terranova builds more proprietary data and execution experience
- Terranova's use of waste biomass as an underground fill lowers cost while adding climate benefits, making the solution cost effective and sustainable
- Federal infrastructure policy, such as the IIJA ($71bn) and FEMA's BRIC ($1bn) program, direct major funding toward flood control and coastal resilience, increasing adoption of Terranova's solution.

Fundraising History

Investors	Capital Raised	Date Raised
Outlander VC, Congruent Ventures, GoAhead Ventures, Gothams, Ponderosa	$7M	Nov 2025

Key Management

Name	Position	Qualifications
Laurence Allen	CEO & Co-Founder	UCB+Stanford; Geotechnical & robotics engineering; leads field operations
Trip Allen	Chairman	25+ years in San Rafael; entrepreneur and climate-tech advocate driving GTM

Sources: [1] Pitchbook [2] Y Combinator [3] LinkedIn [4] MTC [5] FEMA [6] Terranova

Questions?